Machine Learning

CSE 475 | Session 2020-21 | Final exam

Model study answers. Every listed option is addressed. Question titles are paraphrased; exact wording and original diagrams are in the source pages appended to this pack.

Quick revision

Standardization: z = (x - mean) / standard deviation. Fit statistics on training data only.

Gini(D) = 1 - sum(p_k squared). Split impurity = sum((n_j/n) Gini(D_j)).

Perceptron: z = w.x + b; step(z) = 1 for z >= 0, otherwise 0.

Precision = TP/(TP+FP); recall = TP/(TP+FN); accuracy = (TP+TN)/N.

F1 = 2TP/(2TP+FP+FN). Naive Bayes score(c) = P(c) product P(x_j|c).

Iterations/epoch = ceiling(N/batch size), when retaining the last partial batch.

1(a) | What is reinforcement learning?

An agent learns by interacting with an environment. At each step it observes a state, chooses an action, receives a reward and moves to a new state. It learns a policy that maximizes expected cumulative reward, balancing exploration with exploitation. Example: a robot learns a route by receiving rewards for reaching its destination and penalties for collisions.

Revision answer: Learning actions through interaction and reward.

1(b) | What does k represent in k-NN?

k is the number of nearest training samples consulted for a prediction. For classification, their labels vote; for regression, their values are averaged, possibly with distance weighting. Small k can be noisy; large k can blur local patterns. Choose k using validation.

Revision answer: k = number of neighbors used to predict.

1(c) | Linear regression cost function and why

The usual cost is mean squared error: MSE = (1/n) sum(y_i - predicted_y_i)^2. Squaring prevents positive and negative residuals from cancelling and penalizes large errors. For ordinary linear regression, the objective is convex in the coefficients, which supports straightforward optimization. It also corresponds to maximum likelihood under independent Gaussian errors with constant variance.

Revision answer: Usually MSE; it is smooth and convex for ordinary linear regression.

1(d) | Curse of dimensionality

As feature dimension grows, the space expands rapidly, so a fixed dataset becomes sparse. Distances can become less informative and models may need much more data to generalize. Feature selection, regularization and dimensionality reduction can help.

Revision answer: More dimensions make data sparse and learning harder.

1(e) | Limitations of an MLP

A multilayer perceptron may need considerable labeled data, computation and hyperparameter tuning. It can overfit and is difficult to interpret. A plain fully connected MLP does not explicitly exploit spatial or temporal structure as CNNs or sequence models do. The question asks for a limitation; any clearly justified one is sufficient.

Revision answer: MLPs can overfit, require tuning, and lack built-in spatial/temporal structure.

1(f) | Traditional programming versus machine learning

Traditional programming combines explicit rules with input data to produce outputs. Supervised machine learning uses example inputs and target outputs to learn a model, then applies that model to new inputs. Example: hand-written spam rules versus a classifier trained on labeled emails.

Revision answer: Explicit rules are written in programming; useful patterns are learned in ML.

1(g) | Main limitation of ensemble learning

Combining several models usually increases training cost, memory use and prediction latency, and often reduces interpretability. Improvement also depends on useful diversity: highly correlated weak models may add little benefit.

Revision answer: Higher computational cost and lower interpretability are common trade-offs.

2(a) | Sigmoid, ReLU and preference for ReLU

Sigmoid(z) = 1/(1+exp(-z)), with output between 0 and 1. ReLU(z) = max(0,z). Sigmoid saturates for large positive or negative inputs, so its derivative approaches zero. For positive inputs, ReLU has derivative 1, helping gradients propagate and making computation simple. ReLU can still produce dead neurons for persistently negative inputs. Sigmoid remains appropriate for binary classification output probabilities.

Revision answer: ReLU is usually preferred in hidden layers; sigmoid is useful at binary output.

2(b) | How L1 regularization selects features

Minimize prediction loss + lambda sum|w_j|. The absolute-value penalty favors sparse solutions, pushing some coefficients exactly to zero. Features whose coefficients become zero are effectively excluded. Increasing lambda generally increases shrinkage; choose it with validation and scale features when coefficient magnitudes must be comparable.

Revision answer: L1 makes some weights zero, creating a sparse feature set.

2(c) | Cross-validation process and purpose

Purpose: estimate generalization and compare models without repeatedly tuning on the final test set. Divide the training portion into k folds. Train on k-1 folds and validate on the remaining fold, repeat k times, then average the scores. Select hyperparameters and refit on the full training portion. Evaluate once on an untouched test set. Fit preprocessing separately within each training fold to prevent leakage. Diagram: folds 1..k rotate through validation while the other folds train. For time series, use time-respecting splits.

Revision answer: Rotate validation folds, average performance, and keep the final test set untouched.

2(d) | TF-IDF and word relevance

Term frequency measures how often a term occurs in a document, often normalized. Inverse document frequency downweights terms appearing in many documents: a simple form is idf(t)=log(N/df(t)); practical variants use smoothing. TF-IDF = tf(t,d) x idf(t). A frequent word in one document that is rare across the corpus receives a large weight. A ubiquitous word receives a small weight. Example: in a corpus of 100 documents, a term in only 2 documents has idf=log(50).

Revision answer: High local frequency plus low corpus frequency gives high TF-IDF.

2(e) | Outliers and RANSAC

Outliers can distort a fitted line, inflate squared-error loss and shift summary statistics. RANSAC repeatedly samples a minimal subset, fits a candidate model, measures which observations lie within a residual threshold, and retains a model with a strong inlier consensus. It then refits using inliers. Set a suitable threshold and sufficient iterations. It works best when a meaningful inlier model exists.

Revision answer: RANSAC fits a model supported by inliers rather than letting outliers dominate.

2(f) | Feature scaling and standardization

Distance-based models can let a numerically large feature dominate; gradient optimization can also be poorly conditioned. Standardization uses z=(x-mean)/standard deviation. Example: mean height 160 cm, standard deviation 10 cm, height 175 cm gives z=(175-160)/10=1.5. Calculate mean and standard deviation from training data, then reuse them for validation/test. Trees generally do not require scaling for their split decisions.

Revision answer: Standardization expresses features in standard-deviation units.

3(a)(i) | Parametric versus non-parametric models

A parametric model has a fixed finite parameterization chosen before observing the dataset size. Linear regression with a fixed number of features is an example. A non-parametric model has flexibility that can grow with the data, such as k-NN or an unconstrained decision tree. Non-parametric does not mean that no parameters exist. Parametric models may be simpler and data-efficient; non-parametric models can capture more complex patterns but may overfit.

Revision answer: Linear regression is parametric; k-NN and flexible trees are non-parametric.

3(a)(ii) | Explain the sentiment-classification pipeline

Preprocessing: collect student comments about the government decision, assign sentiment labels, remove duplicates, split data, and turn text into features such as TF-IDF. Learning: train a classifier on training features and labels. Evaluation: predict held-out comments and compare with their true labels using suitable metrics. Prediction: apply the same fitted preprocessing and final model to new comments. In the scanned diagram, the labels in evaluation are the ground truth for comparison; do not feed test labels into training.

Revision answer: Labeled comments -> preprocessing/split -> training -> held-out evaluation -> new-comment prediction.

3(b)(i) | Gini impurity: Weather versus Temperature

There are 8 examples: 4 Yes and 4 No. Parent Gini = 1-(4/8)^2-(4/8)^2 = 0.5.

Weather: Sunny has 3 No, so Gini=0. Overcast has 2 Yes, so Gini=0. Rain has 2 Yes and 1 No, so Gini=1-(2/3)^2-(1/3)^2=4/9.

Weighted Weather impurity = (3/8)0+(2/8)0+(3/8)(4/9)=1/6, about 0.1667. Gain=0.5-1/6=1/3.

Temperature: Hot has 1 Yes and 2 No, Gini=4/9. Mild has 1 Yes and 1 No, Gini=1/2. Cool has 2 Yes and 1 No, Gini=4/9.

Weighted Temperature impurity=(3/8)(4/9)+(2/8)(1/2)+(3/8)(4/9)=11/24, about 0.4583. Gain=1/24.

Using the multiway categorical split requested by this table, choose Weather because it leaves less impurity. A binary-split implementation must evaluate category partitions separately.

Revision answer: Choose Weather: impurity 1/6 versus 11/24.

3(b)(ii) | How boosting works

Boosting builds an ensemble sequentially. Each new learner focuses on what the current ensemble handles poorly. AdaBoost increases the influence of misclassified samples; gradient boosting fits the negative gradient of the current loss, often described as fitting residual errors. Combine learners by weighted voting or addition. Learning rate and stopping control overfitting.

Revision answer: Sequential learners correct earlier errors and combine into a stronger predictor.

3(c)(i) | Steps for training a neural network

Prepare and split the data; scale inputs where appropriate. Choose architecture, activations, loss and optimizer. Initialize weights. For each mini-batch, perform a forward pass, compute loss, backpropagate gradients and update parameters. Repeat across epochs, monitor validation loss, tune with validation data and optionally use early stopping. Finally evaluate once on held-out test data.

Revision answer: Forward pass -> loss -> backpropagation -> update; repeat and validate.

3(c)(ii) | Keras code and replacing sigmoid with softmax

Sequential stacks layers in order. Dense(16, activation="relu") creates a fully connected hidden layer with 16 units; the next Dense(16, ...) creates another. Dense(1, activation="sigmoid") creates one output between 0 and 1, suitable for binary classification with an appropriate loss. Its trainable parameter count depends on the unspecified input dimension d: (d+1)16 + (16+1)16 + (16+1)1 = 16d+305. Replacing sigmoid with softmax while retaining one output makes softmax always equal exp(z)/exp(z)=1, so it cannot distinguish classes. For mutually exclusive multiclass classification use multiple softmax outputs and a corresponding categorical loss.

Revision answer: A one-unit softmax always returns 1; retain sigmoid for a single binary output.

4(a) | Vanishing gradient

During backpropagation, many small derivatives are multiplied through layers. The resulting gradients in early layers can become tiny, so those weights learn slowly. Saturating activations worsen this. Appropriate initialization, non-saturating activations, normalization and residual connections can help.

Revision answer: Early layers learn slowly when backpropagated gradients shrink toward zero.

4(b) | Transfer learning

Reuse knowledge from a model trained on a source task for a related target task. For example, use a pretrained image encoder, train a new classifier head and optionally fine-tune selected layers. It can reduce target-data and training requirements; unrelated source data can cause negative transfer.

Revision answer: Adapt a pretrained model to a new task.

4(c) | Why accuracy is insufficient for imbalanced data

If 99 of 100 cases are negative, a model that always predicts negative has 99% accuracy while detecting no positives. Inspect the confusion matrix, precision, recall, F1 and appropriate ranking metrics according to the cost of errors.

Revision answer: High accuracy can hide failure on the minority class.

4(d) | Why differentiable activations?

Gradient-based learning uses derivatives through the chain rule to determine how changing weights changes loss. Activations should therefore be differentiable almost everywhere or have usable subgradients. ReLU is not differentiable at zero, but implementations choose a subgradient there, so it remains trainable.

Revision answer: Backpropagation needs derivatives or usable subgradients.

4(e) | Metric for continuous-valued predictions

Use a regression metric such as MAE, MSE or RMSE. MAE averages absolute errors and is in the target unit. RMSE is also in the target unit and emphasizes large errors. Select according to the intended error cost.

Revision answer: MAE, MSE or RMSE are suitable regression metrics.

4(f) | What are outliers?

Observations unusually far from the typical pattern. They may reflect measurement errors, rare valid events or a different data-generating process. Investigate before deleting them; automatic removal can discard important cases.

Revision answer: Outliers are atypical observations, not necessarily errors.

4(g) | Hyperparameters and examples

Hyperparameters are settings chosen outside the main parameter-fitting process. Examples: k in k-NN, tree maximum depth, neural-network learning rate, regularization strength and batch size. Tune them on validation data rather than the final test set.

Revision answer: Hyperparameters control the learning setup; weights are learned parameters.

5(a) | Perceptron, neuron analogy and numerical output

A biological neuron integrates signals and may fire; a perceptron computes a weighted sum and applies an activation. Inputs resemble incoming signals, weights resemble synaptic strengths, and the bias shifts the firing threshold. Given x=[1,-2], w=[0.5,0.25], b=1: z=(1)(0.5)+(-2)(0.25)+1 = 0.5-0.5+1=1. With step(z)=1 for z>=0, output=1. The analogy is simplified; a perceptron is not a complete biological model.

Revision answer: Net input 1; step-activation output 1.

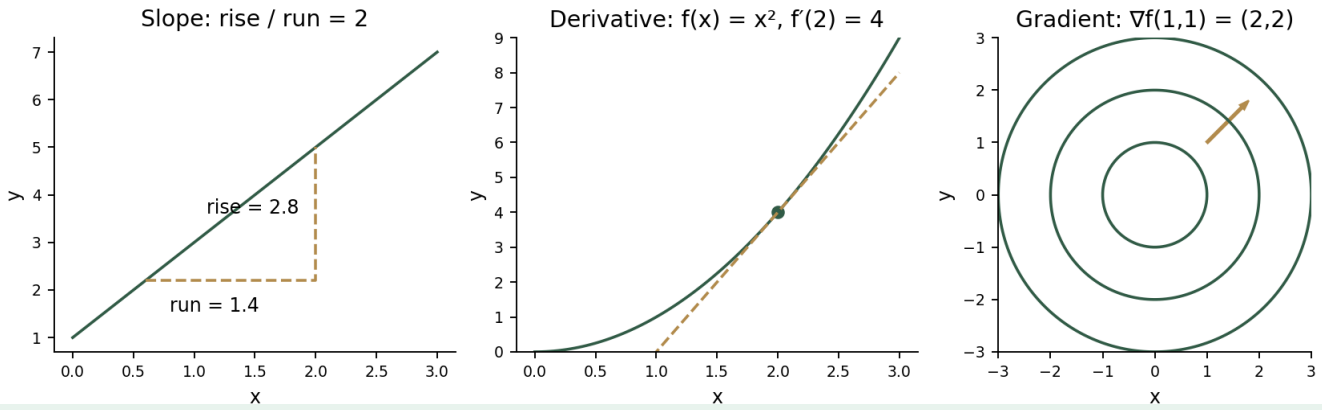
5(b) | Epoch, batch and iterations

An epoch is one complete pass through the training set. A batch is a subset used for one update. Divide 60,000 by 128: 468 full batches cover 59,904 examples, leaving 96. Keeping that final partial batch gives 469 iterations per epoch. If an implementation explicitly drops the partial batch, it performs 468 and omits those examples in that pass.

Revision answer: 469 iterations when the final partial batch is retained.

5(c) | Slope, derivative and gradient

Slope of a straight line is rise/run, Delta y/Delta x. A derivative is the limiting local slope of a one-variable function: f'(x)=lim[h->0](f(x+h)-f(x))/h. For f(x)=x^2, f'(2)=4, the slope of its tangent at x=2. A gradient collects partial derivatives for a multivariable function. For f(x,y)=x^2+y^2, gradient=(2x,2y); at (1,2) it is (2,4). Draw a line with a rise/run triangle, a curve with its tangent, and contour circles with an outward gradient arrow. The site supplies an explanatory figure below this answer.



Revision answer: Slope: line steepness; derivative: local slope; gradient: vector of partial derivatives.

5(d) | k-means limitations and k-means++

k-means requires k in advance, is sensitive to initialization and outliers, converges to local optima and favors roughly compact, similarly scaled clusters under Euclidean distance. k-means++ chooses the first center randomly, then chooses later centers with probability proportional to squared distance from the nearest selected center. This spreads initial centers and often improves convergence and solution quality. It does not solve the choice of k, non-spherical clusters or outlier sensitivity.

Revision answer: k-means++ improves initialization, not every limitation of k-means.

5(e) | Why k-NN is lazy; handling ties

k-NN stores the training examples and defers most modeling work until a query arrives, hence lazy learning. Resolve voting ties using a declared rule: distance-weighted voting, the class of the nearest tied neighbor, or a deterministic class order. Odd k reduces binary-class vote ties but does not eliminate multiclass ties or equal-distance ambiguities.

Revision answer: It postpones work until prediction; use an explicit tie-breaking rule.

5(f) | Feature importance and how to measure it

Feature importance describes how strongly a feature contributes to predictions or predictive performance. Methods include permutation importance on held-out data, standardized coefficients for suitable linear models, and tree impurity reductions. Permuting a useful feature generally reduces performance. Correlated features can share or hide importance; impurity-based importance can favor high-cardinality features. Importance is not proof of causation.

Revision answer: Measure contribution with held-out permutation tests or model-specific methods.

6(a)(i) | What a confusion matrix represents

It cross-tabulates actual and predicted class labels. For binary classification it contains TP (positive correctly predicted), FN (positive missed), FP (negative incorrectly predicted positive) and TN (negative correctly predicted). State the row/column convention because libraries differ.

Revision answer: A confusion matrix counts each type of correct and incorrect prediction.

6(a)(ii) | Accuracy, precision, recall and F1 calculation

Actual positives=100; actual negatives=50. TP=90, FN=100-90=10, TN=30, FP=50-30=20. Total=150. Accuracy=(90+30)/150=0.80=80%. Precision=90/(90+20)=9/11, about 81.82%. Recall=90/(90+10)=0.90=90%. F1=2(90)/(180+20+10)=180/210=6/7, about 85.71%. F1 balances precision and recall without the often-dominant TN count, making it informative for a positive minority class, although it ignores TN and is not universally the best metric.

Revision answer: Accuracy 80%; precision 81.82%; recall 90%; F1 85.71% (rounded).

6(b)(i) | SVM kernel trick

A nonlinear feature map phi(x) may make data linearly separable in a higher-dimensional space. A kernel computes K(x,z)=phi(x).phi(z) without explicitly constructing that map. The SVM can optimize its margin using kernel values. Examples include polynomial and radial-basis-function kernels. Tune kernel parameters and regularization on validation data; a higher-dimensional representation does not guarantee good generalization.

Revision answer: Kernels compute feature-space inner products implicitly.

6(b)(ii) | Identify the three fits and reduce overfitting

Reading the scanned figures from left to right: the smooth curved separator is the intended good fit, the straight separator is underfitting, and the highly wiggly separator is overfitting. Underfitting misses real structure; overfitting follows individual points or noise and generalizes poorly. Reduce overfitting with regularization, a simpler model, more representative data, early stopping, pruning or suitable augmentation. Compare training and validation performance to substantiate the visual diagnosis.

Revision answer: Left: good fit. Middle: underfit. Right: overfit, as intended by the sketch.

6(c)(i) | Activity data: prior probabilities

Count all 10 rows in the original table. Play occurs in rows 1,3,4,6,9: 5 times. Study occurs in rows 2,7,10: 3 times. Sleep occurs in rows 5,8: 2 times. Therefore P(Play)=5/10=0.5; P(Study)=3/10=0.3; P(Sleep)=2/10=0.2.

Revision answer: Play 0.5; Study 0.3; Sleep 0.2.

6(c)(ii) | Naive Bayes: Near, Bad weather, Off mood

Use score(c)=P(c)P(Near|c)P(Bad|c)P(Off|c), assuming conditional independence. For Play: 0.5 x (2/5) x (0/5) x (5/5)=0. For Study: 0.3 x (1/3) x (3/3) x (1/3)=1/30. For Sleep: 0.2 x (1/2) x (2/2) x (1/2)=1/20. Normalize: total=1/12, so posterior Play=0, Study=0.4, Sleep=0.6. Predict Sleep. With add-one smoothing across 3 deadline, 2 weather and 2 mood categories, scores are 9/392, 4/125 and 3/100 respectively; Study then narrowly wins. The answer therefore depends on the declared smoothing convention; the unsmoothed classroom calculation gives Sleep. Smoothing avoids a zero likelihood for an unseen feature/class combination.

Revision answer: Predict Sleep; unsmoothed posterior is 0.60.

Reference reading

Christopher M. Bishop, Pattern Recognition and Machine Learning.

Aurelien Geron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow.

scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html

These are supporting reading references, not claims that this paper or its answer key appears in those books. Course-specific terminology and source ambiguities are noted beside the affected answers.

Shahjalal University of Science and Technology
Department of Computer Science and Engineering
4th Year 1st Semester Final Examination, 2024 (Session: 2020-21)
Course Code: CSE 475 Course Title: Machine Learning
Time: 3.0 hrs **Credits: 3.0** **Total Marks: 100**

Group A
[Answer all questions]

1. Answer any FIVE. **5x2=10**

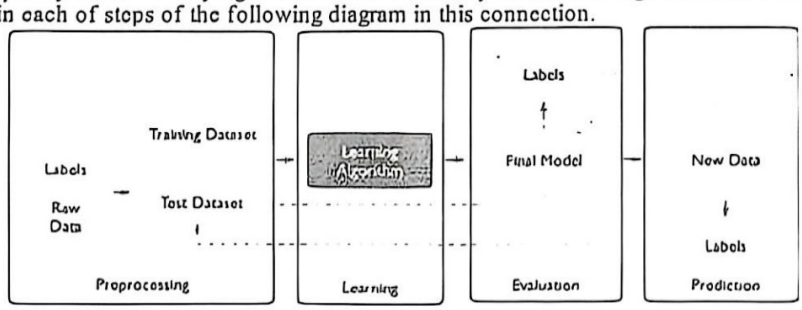
a) What is reinforcement learning?
b) What does the "k" represent in k-NN?
c) Which cost function is used in linear regression? Why?
d) What does "curse of dimensionality" mean?
e) What is the limitation of an MLP model?
f) What is the difference between traditional programming and machine learning?
g) What is the main limitation of ensemble learning?

2. Answer any FOUR. **4x5=20**

a) Define sigmoid and ReLU activation function. Why is ReLU preferred over sigmoid function in neural networks?
b) Explain how L1 regularization performs feature selection.
c) What is the purpose of cross-validation? Explain the underlying process with appropriate diagram.
d) Define the term TF-IDF. How does TF-IDF assess the word relevancy in encoding?
e) What happens when there are outliers in the dataset? Explain the RANSAC algorithm for handling outliers.
f) Why is feature scaling important? Explain the standardization method with an example.

3. Answer any TWO. **2x10=20**

a) i. Differentiate between parametric and non-parametric ML models. Give examples of each type. **3+7**
ii. Suppose you are classifying sentiments of university students on a government's decision. Explain each of steps of the following diagram in this connection.



b) i. Consider the following dataset with features Weather (Sunny, Overcast, Rain) and Temperature (Hot, Mild, Cool) along with the target class (Yes/No). Using the Gini impurity criterion, determine which feature should be chosen for the first split in a decision tree. **6+4**

Instance	Weather	Temperature	Class
1	Sunny	Hot	No
2	Sunny	Hot	No
3	Overcast	Hot	Yes
4	Rain	Mild	Yes
5	Rain	Cool	Yes
6	Rain	Cool	No
7	Overcast	Cool	Yes
8	Sunny	Mild	No

ii. Explain how the boosting approach of an ensemble works?

c) i. Briefly explain the steps for training a neural network. **5+5**
ii. Explain the meaning of the following piece of Python code.

```
model = keras.Sequential([
    layers.Dense(16, activation="relu"),
    layers.Dense(16, activation="relu"),
    layers.Dense(1, activation="sigmoid")
])
```

What is the effect of replacing sigmoid with softmax in a perceptron?

CamScanner

Group B

[Answer all questions]

4. Answer any FIVE. 5x2 = 10

a) What is the vanishing gradient problem?
b) What is transfer learning?
c) Give one reason why accuracy is not enough for imbalanced datasets.
d) Why should the activation function in ANN be differentiable?
e) Which metric can be used to evaluate a model that outputs a continuous value?
f) What are outliers?
g) What is a hyperparameter? Give suitable examples.

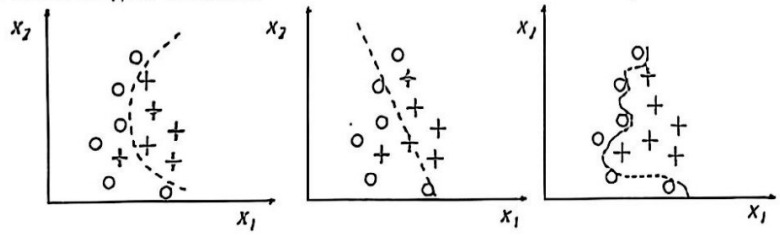
5. Answer any FOUR. 4x5 = 20

a) How is perceptron related with the biological neuron? Given the input vector [1, –2], the weight vector [0.5, 0.25], and a bias of 1, compute the output of a perceptron using a step activation function.
b) Define epoch and batch in the context of ML. If batch size = 128 and dataset has 60000 samples, how many iterations per epoch are required?
c) Define slope, derivative, and gradient? Briefly explain each with figures.
d) What are the limitations of the k-means algorithm? How does the k-means++ algorithm address them?
e) Why is KNN said to be a lazy learner? What happens when the majority voting in KNN results in a tie?
f) What is meant by feature importance? How can it be determined?

6. Answer any TWO. 2x10=20

a) **i.** What does the confusion matrix represent? 3 + 7
ii. A dataset has 100 positives and 50 negatives. A classifier correctly predicts 90 positives and 30 negatives. Compute accuracy, precision, recall, and F1-score. Why is the F1-score useful for imbalanced datasets?

b) **i.** Explain the kernel-trick in SVM to find separating hyperplanes in a high-dimensional space. 5 + 5
ii. Explain which of the following cases represent underfitting, overfitting, and a good fitting. How can overfitting be reduced?



g) Consider your activities of last days: 4+6

Deadline	Weather	Mood	Activity
Urgent	Good	Off	Play
Urgent	Bad	Off	Study
Near	Good	Off	Play
None	Good	Off	Play
None	Bad	On	Sleep
None	Good	Off	Play
Near	Bad	On	Study
Near	Bad	Off	Sleep
Near	Good	Off	Play
Urgent	Bad	On	Study

i. Calculate the prior probabilities: P(Play), P(Study), P(Sleep)
ii. Given that there is a deadline near, the weather is bad and mood is off, find out what you will do today? Use Naive Bayes' classifier

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